



University of Al-Ameed
College of Medicine
Department of Anatomy



Lecture 2

SCALP AND FACE

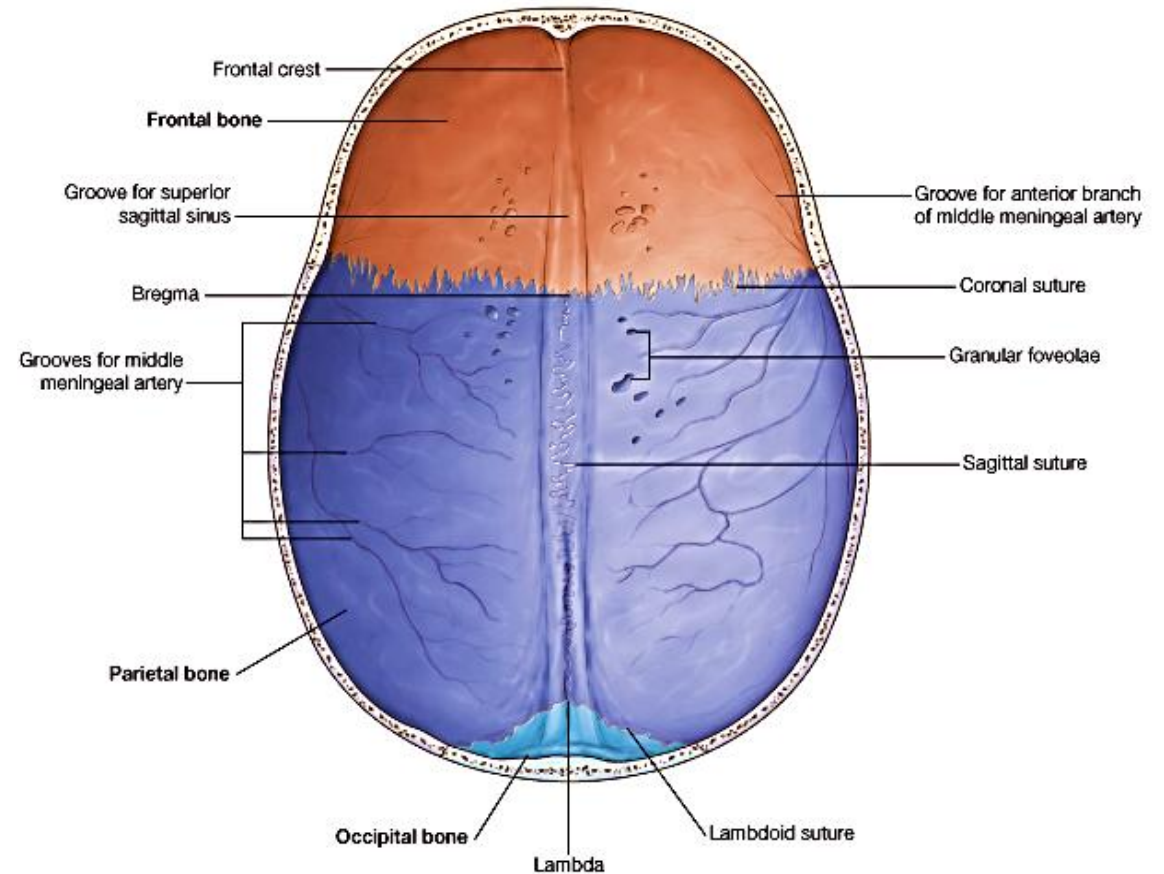
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Objectives

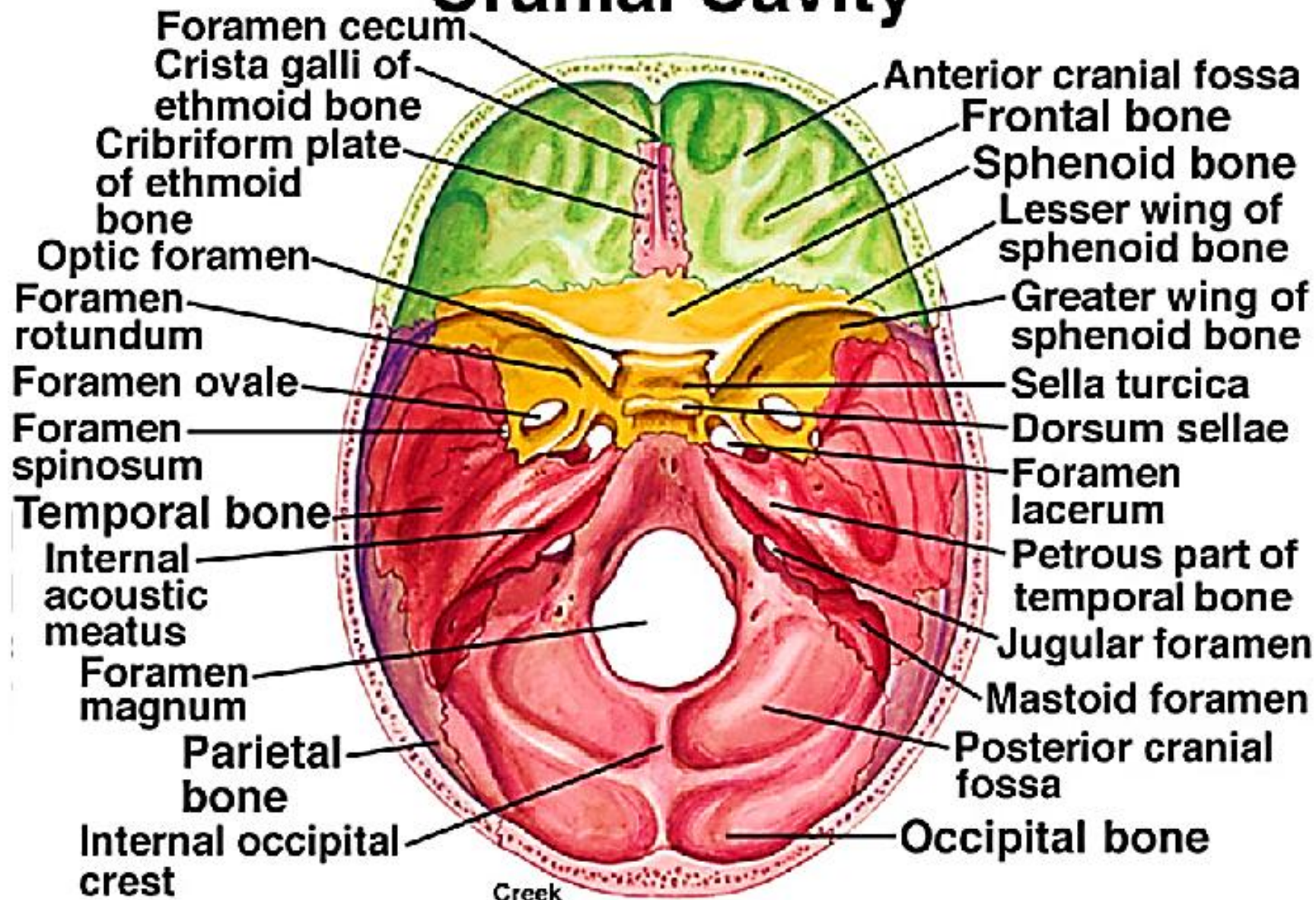
- Illustrate the anatomical structure of cranial cavity.
- Describe anatomy of the SCALP with its innervation and blood supply.
- Define the FACE and facial muscles with their innervation and blood supply.

CRANIAL CAVITY

- It is the space within the calvaria that contains the brain, meninges, proximal parts of the cranial nerves, blood vessels, and cranial venous sinuses.
- **Roof :**
- Sutures
- Frontal crest
- Vascular grooves

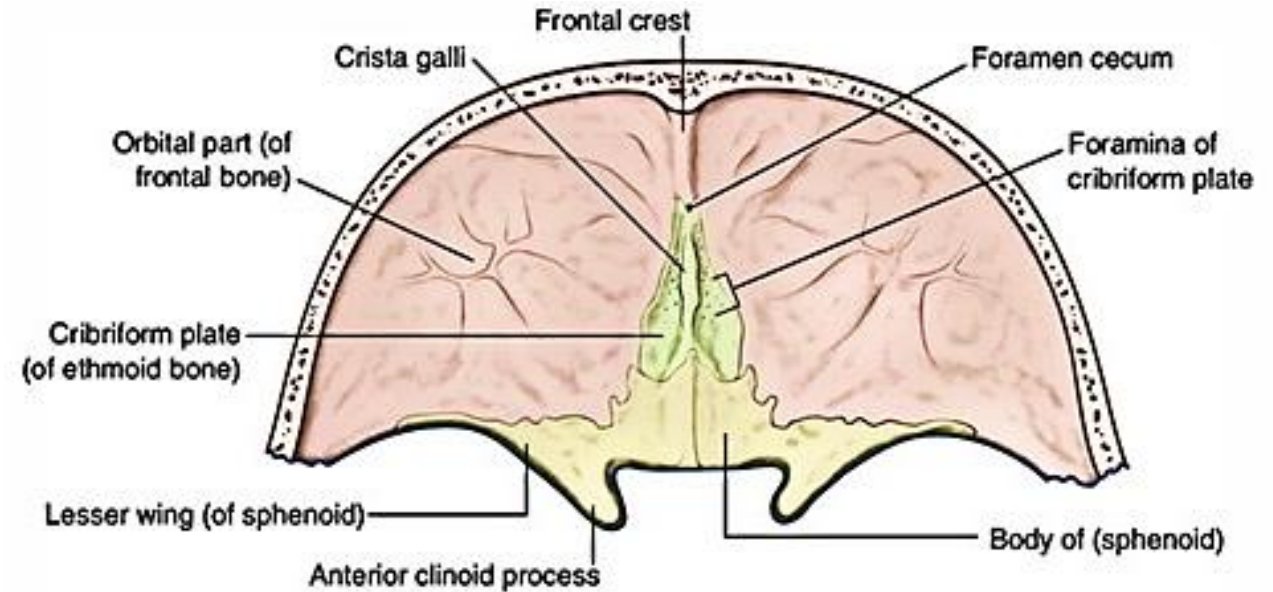


Cranial Cavity



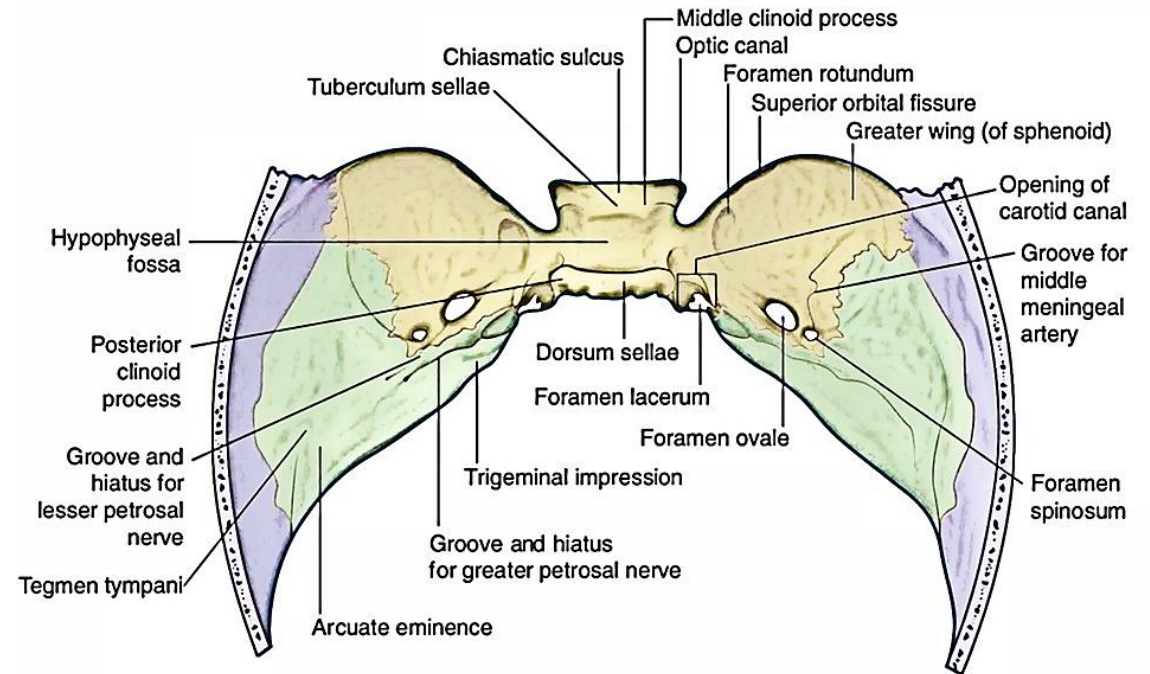
Anterior Cranial Fossa

- It is above the nasal cavity and the orbits, and it is filled by the frontal lobes of the cerebral hemispheres.
- Frontal crest
- Foramen cecum
- Crista galli
- Cribriform plate of the ethmoid
- Anterior clinoid processes



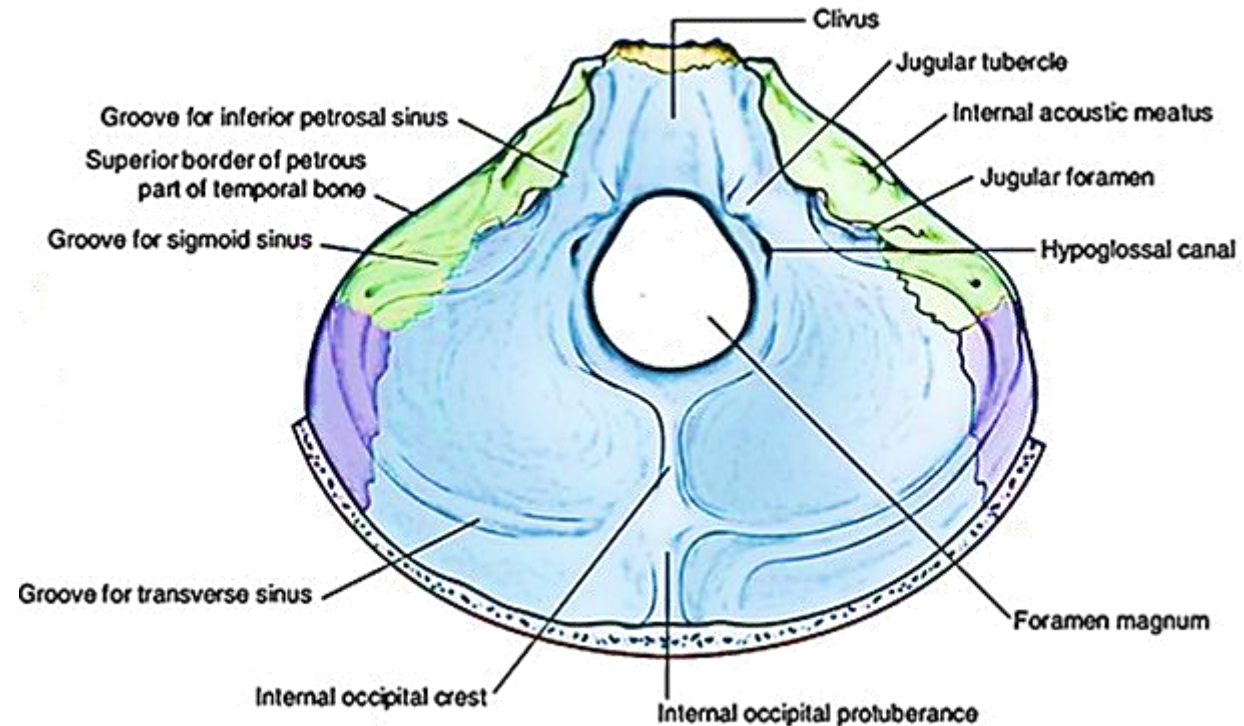
Middle Cranial Fossa

- The boundary between the anterior and middle cranial fossae in the midline is the anterior edge of the chiasmatic sulcus, which is a smooth groove stretching between the optic canals across the body of sphenoid.
- The posterior boundaries of the middle cranial fossa are formed by the anterior surface of the petrous and mastoid part of temporal bone.
- Sella turcica (hypophysial fossa)
- Middle and posterior clinoid processes
- Superior orbital fissure
- Foramen rotundum, ovale, spinosum, carotid canal and foramen lacerum
- Tegmen tympani



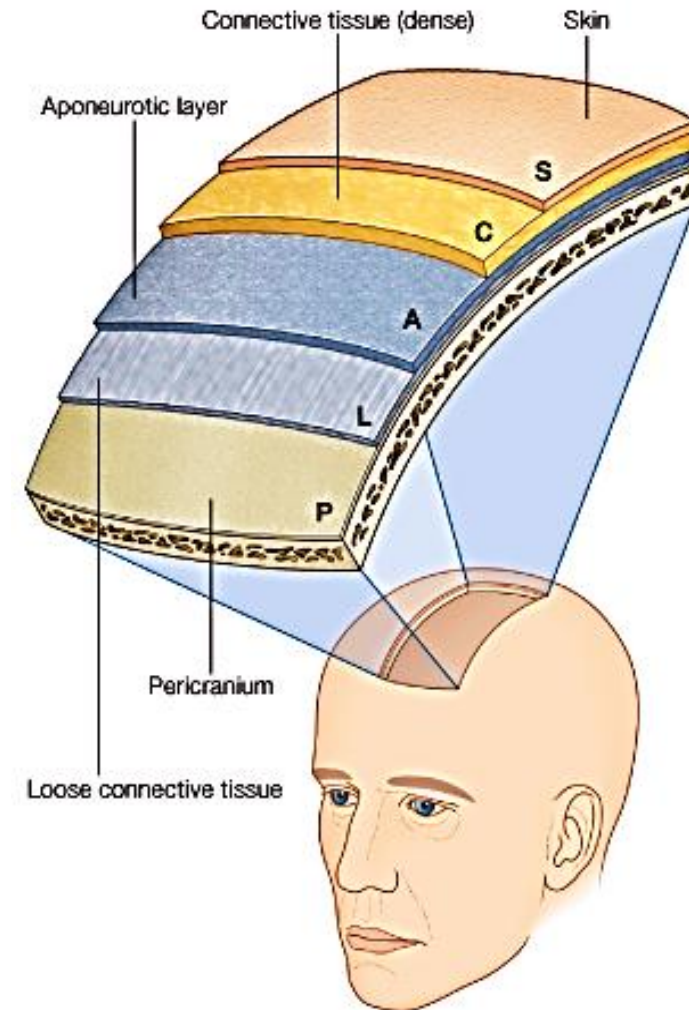
Posterior Cranial Fossa

- It is the largest and deepest of the three cranial fossae and contains the brainstem (midbrain, pons, and medulla) and the cerebellum.
- The anterior boundaries of the posterior cranial fossa in the midline are the dorsum sellae and the clivus. Posteriorly the squamous part of the occipital bone.
- Foramen magnum
- Internal acoustic meatus
- Jugular foramen and tubercle
- Hypoglossal canal
- Condylar canal



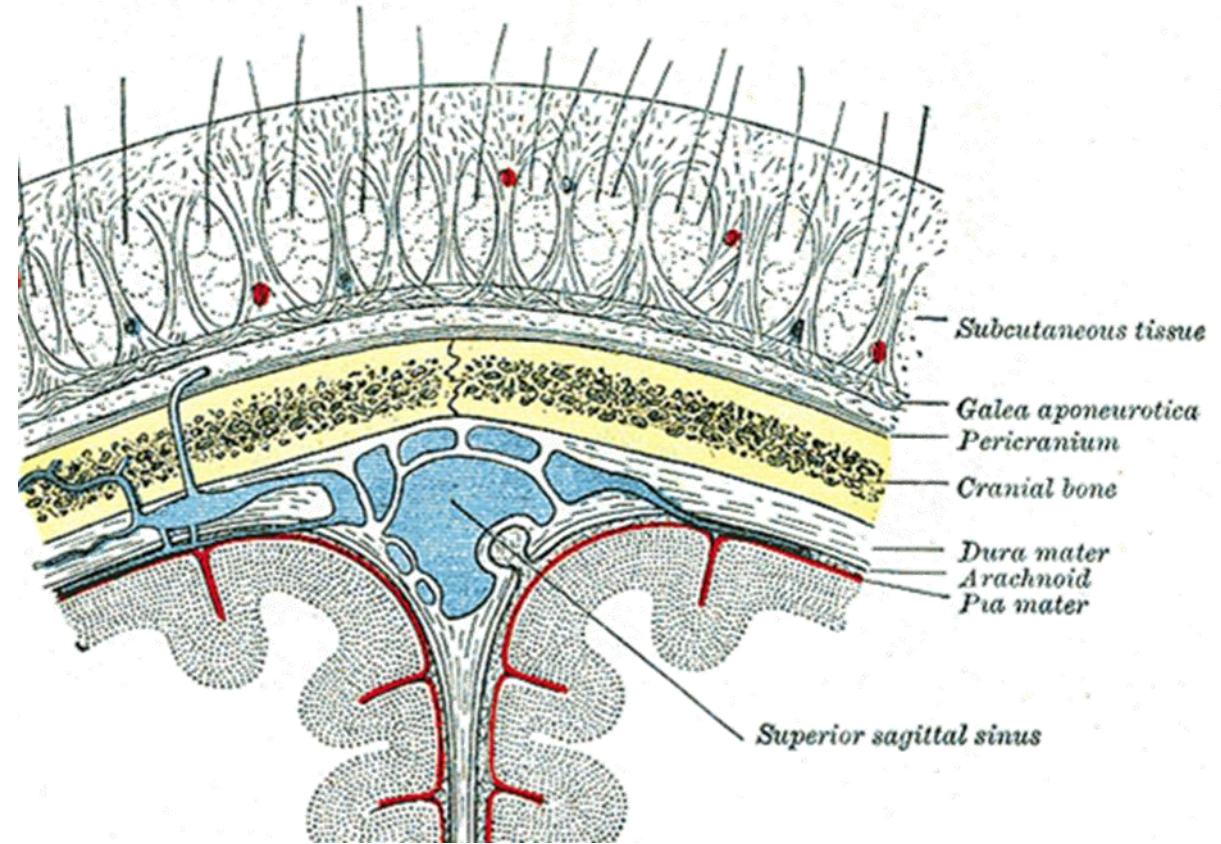
SCALP

- It is the part of the head that extends from the superciliary arches anteriorly to the external occipital protuberance and superior nuchal lines posteriorly. Laterally it continues inferiorly to the zygomatic arch.
- The scalp is a multilayered structure with layers that can be defined by the word itself:
 1. S- skin
 2. C- connective tissue (dense)
 3. A- aponeurotic layer
 4. L- loose connective tissue
 5. P- pericranium



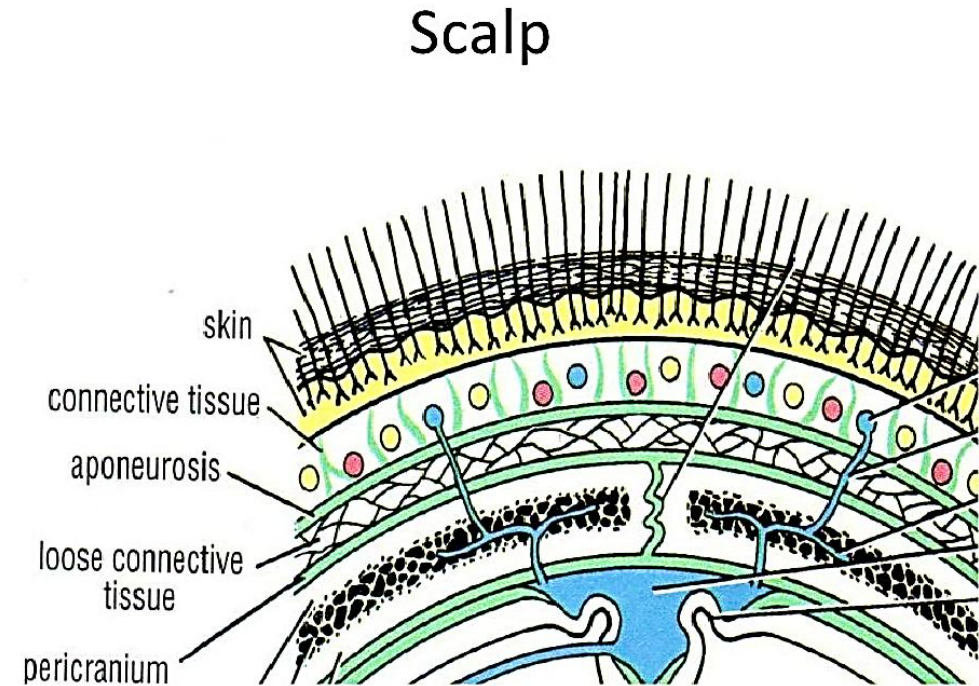
Scalp Proper

- It is the first three layers that are tightly held together to form a single unit.
- It is the tissue torn away during serious scalping injuries.
- **The skin** is thick (among the thickest in the body, measuring 3 to 8 mm.), hairy with numerous sebaceous and sweat glands.
- It is connected to the galea aponeurotica via numerous vertical septa that minimize shearing of the skin on the skull.
- They move with the scalp.



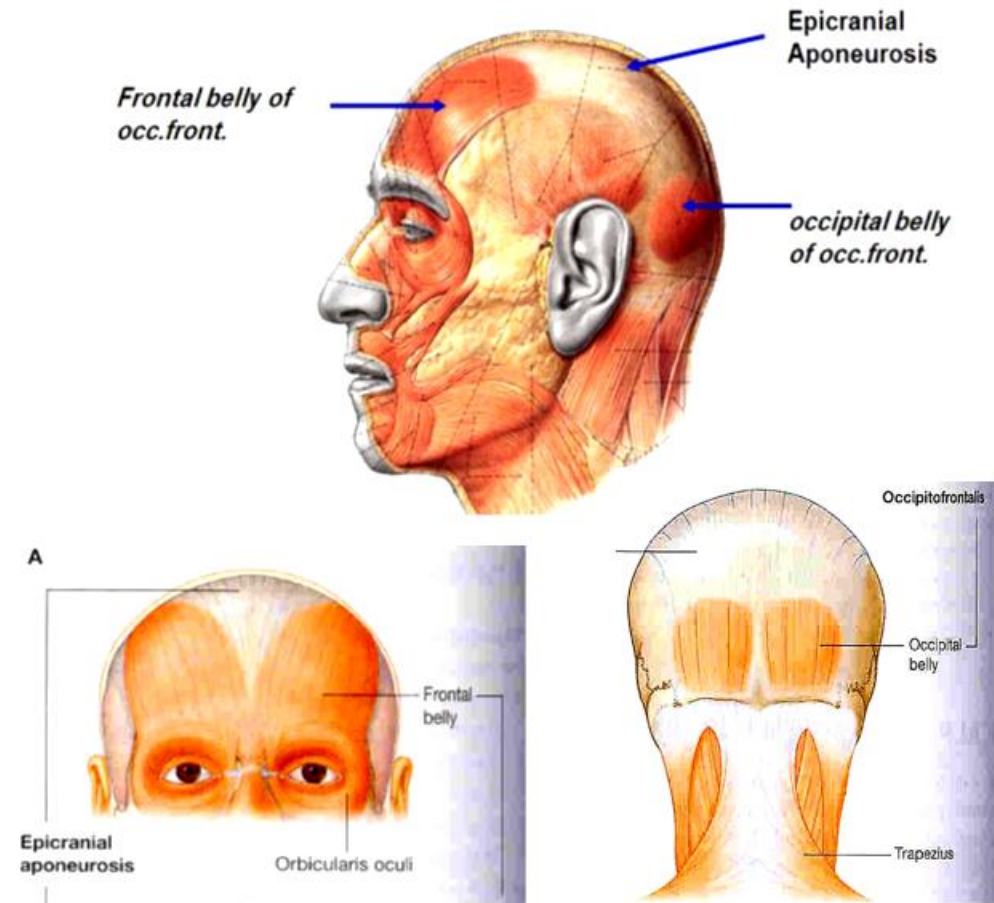
Scalp Proper

- **Connective Tissue** is a fibro-fatty layer which is adherent to the skin and to the underlying aponeurosis by fibrous septa.
- It contains dense connective and fatty tissue and hair follicles, along with veins, arteries, lymphatics, and sensory nerves of the scalp.
- **Aponeurotic Layer** (Galea aponeurotica) is a thin and tendinous sheet that unites the frontal and occipital bellies of occipitofrontalis muscle.
- It is attached laterally to the temporal fascia. The galea is also attached with the subcutaneous musculoaponeurotic system (SMAS) of the face.



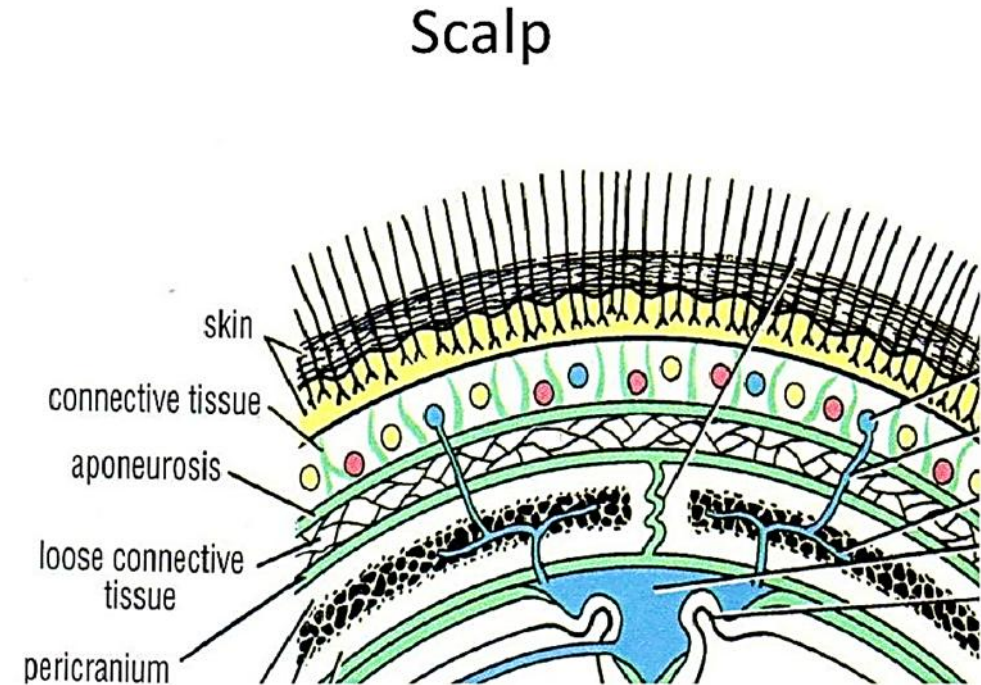
Occipitofrontalis Muscle

- It has a frontal belly anteriorly, and occipital belly posteriorly, and an aponeurotic tendon (galea aponeurotica) connecting the two bellies.
- **Frontal Belly** arises from the anterior part of the aponeurosis and inserted into the skin of the eye brows (has no bony attachment). It blends with orbicularis oculi muscle.
- It elevates the eyebrows giving the face a surprised looking and produces transverse wrinkles of the forehead.
- **Occipital Belly** arises from the lateral two thirds of the superior nuchal line of the occipital bone and from the mastoid part of the temporal bone.



Loose Areolar Tissue

- It occupies the subaponeurotic space.
- It contains few arteries and the important emissary veins. This layer is easily dissected and is often the plane of cleavage in avulsion or “scalping” injuries.
- The (4th) layer of the scalp is the **Dangerous Layer** because pus or blood spreads easily in it.
- Infection in this layer can spread into the bones through the diploic veins causing osteomyelitis.
- It can spread through the emissary veins to the intracranial venous sinuses to cause Venous Sinus thrombosis.



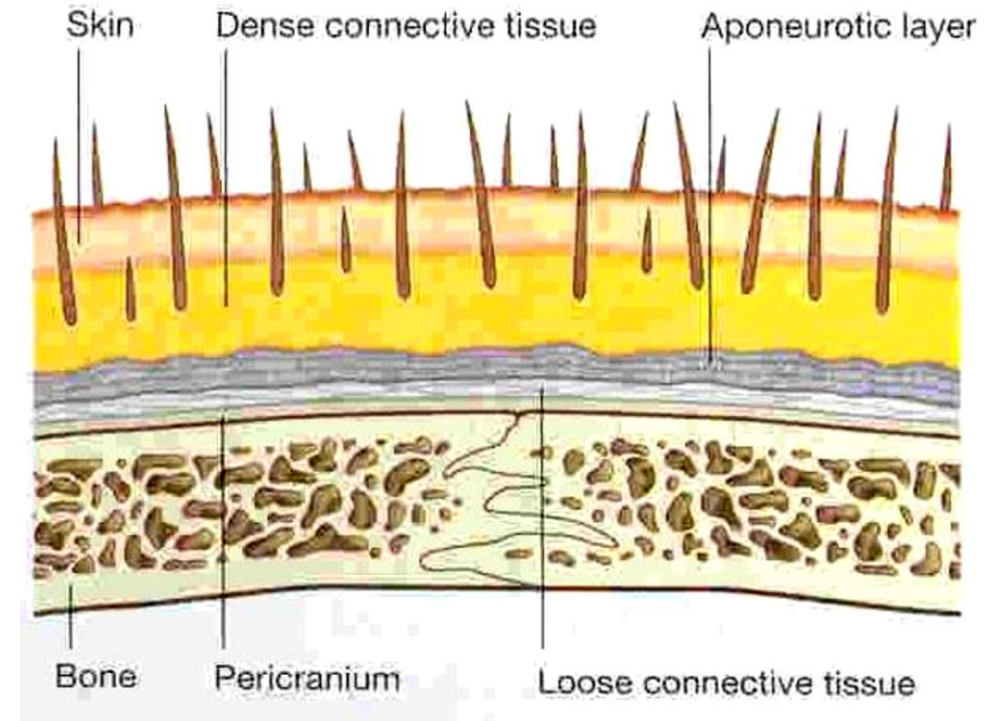
Dangerous Layer

- An infection or fluid can spread only into the eye lids and the root of the nose because of the attachment of the frontalis into the skin and not to the bone.



Pericranium

- It is the deepest layer of the scalp.
- It is the periosteum on the outer surface of the calvaria, but is removable.
- It is distinctly separate from the innominate fascia above it.
- At the sutures it becomes continuous with the periosteum on the outer surface of the bones.



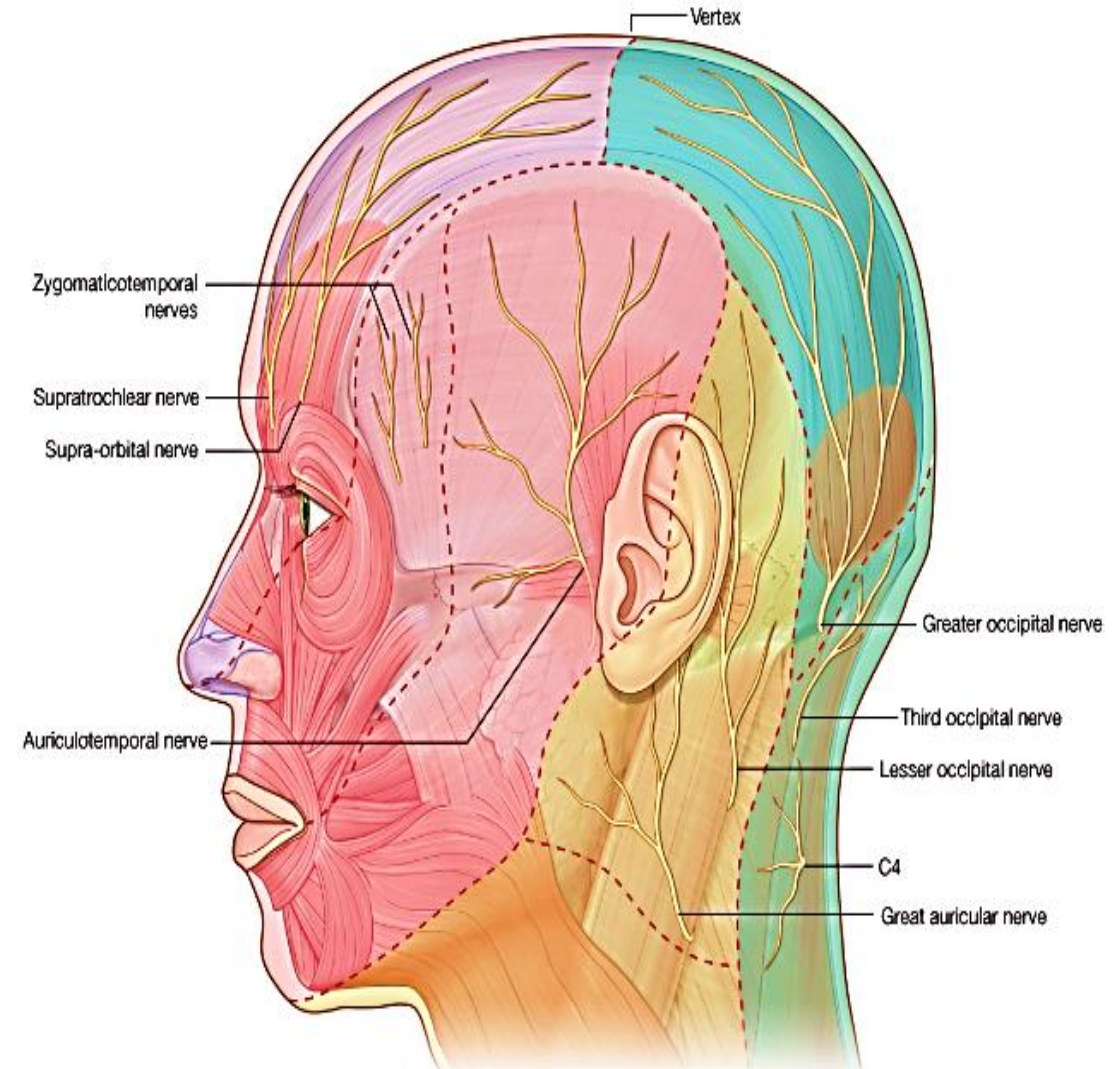
Innervation of Scalp

- **Anterior to the ears and the vertex:**

1. Supratrochlear nerve
2. Supra-orbital nerve
3. Zygomaticotemporal nerve
4. Auriculotemporal nerve

- **Posterior to the ears and the vertex:**

1. Great auricular nerve
2. Lesser occipital nerve
3. Greater occipital nerve
4. Third occipital nerve



Blood vessels of Scalp

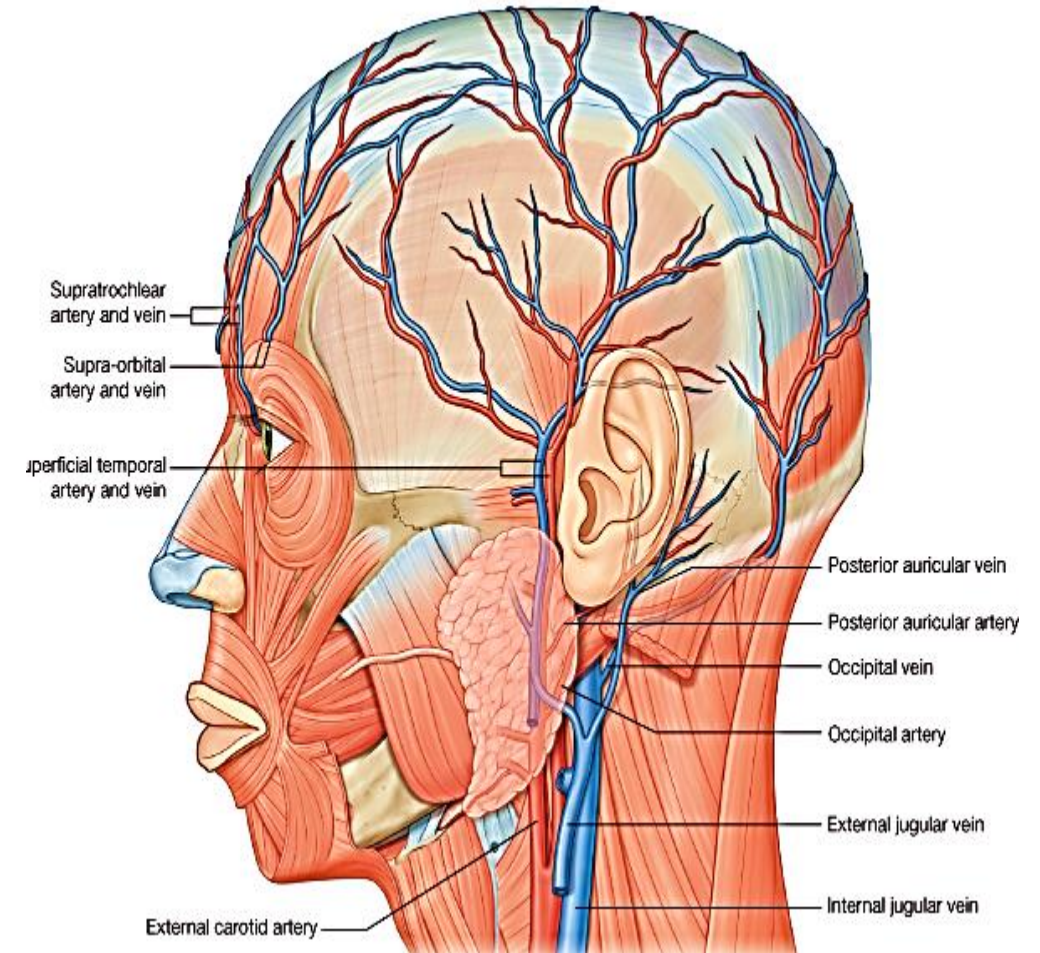
A. Branches from the ophthalmic artery:

- The supratrochlear A
- The supra-orbital A

B. Branches from the external carotid artery:

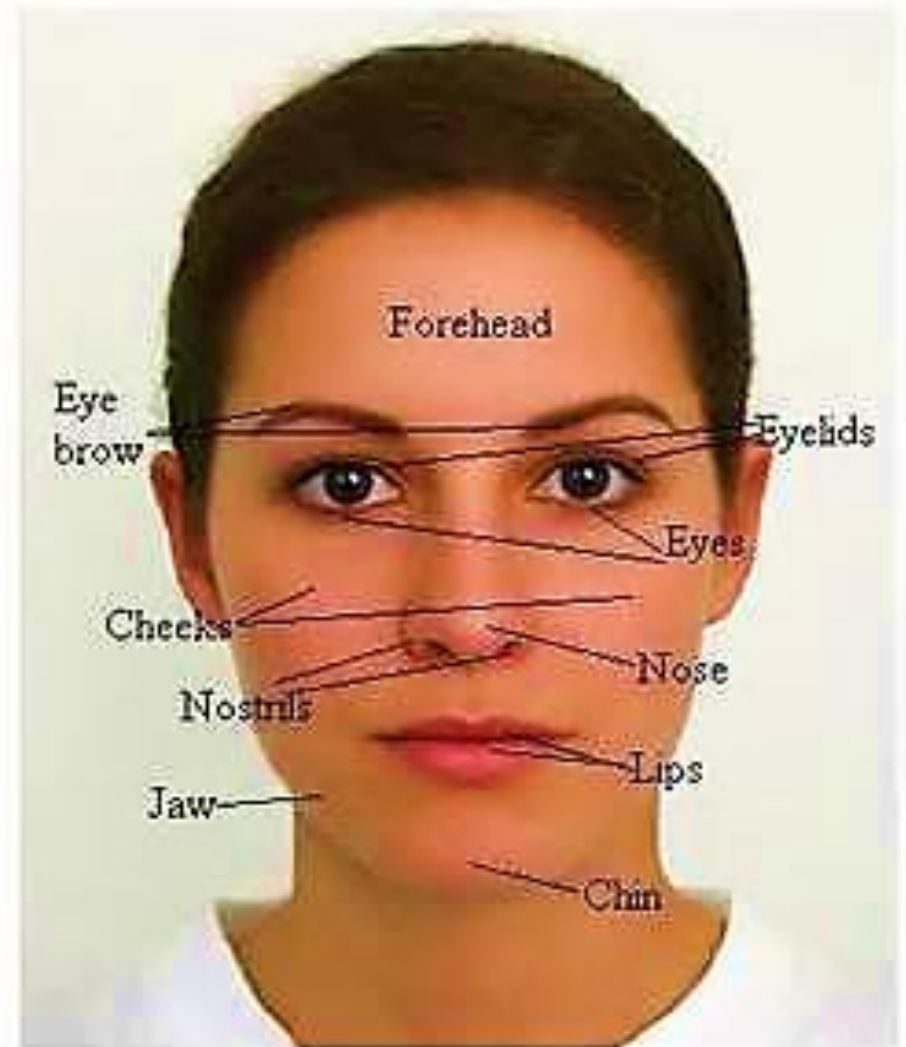
- Posterior auricular artery
- Occipital artery
- Superficial temporal artery

❖ *Veins draining the scalp follow a pattern similar to the arteries.*



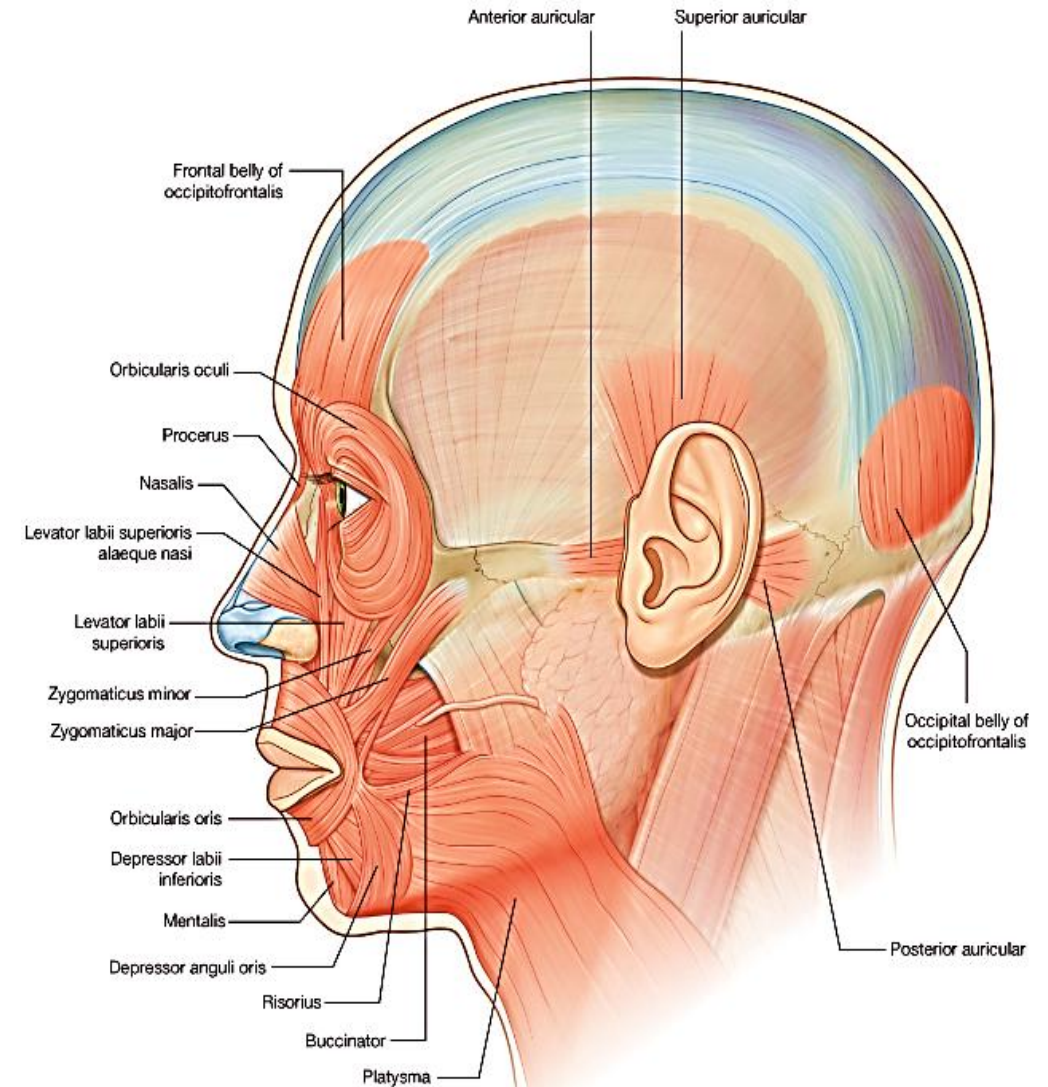
FACE

- It is the region between the superciliary arches superiorly, the lower edge of the mandible inferiorly, and as far back as the ears on either side.

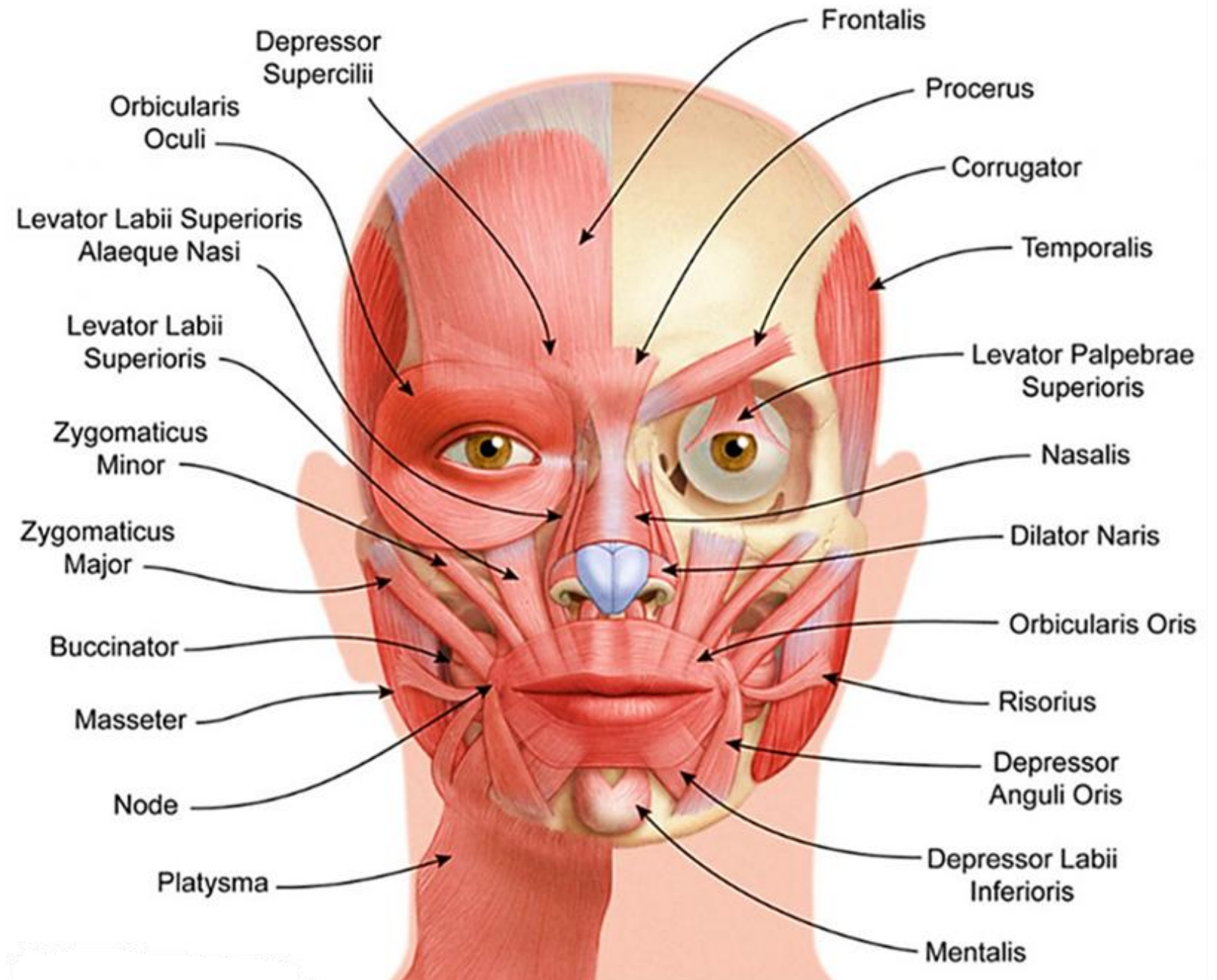


FACIAL MUSCLES

- They are in the superficial fascia, with origins from either bone or fascia, and insertions into the skin.
- Because these muscles control expressions of the face they are sometimes referred to as muscles of 'facial expression'.
- They also act as sphincters and dilators of the orifices of the face (i.e. the orbits, nose, and mouth).
- All muscles of facial expression are supplied by Facial nerve.



FACIAL MUSCLES

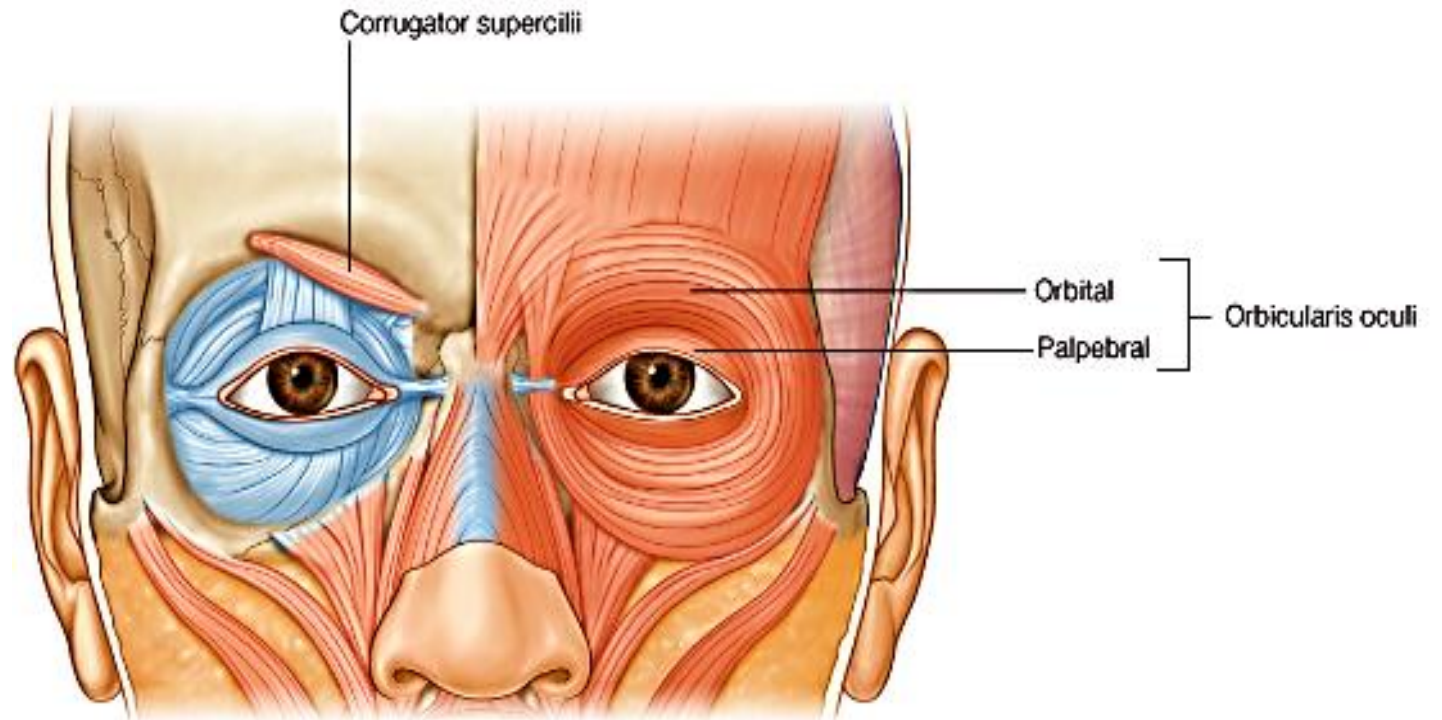


FACIAL MUSCLES

- **Orbital group:**

Two muscles are associated with the orbital group:

1. **Orbicularis oculi**
2. **Corrugator supercilii.**

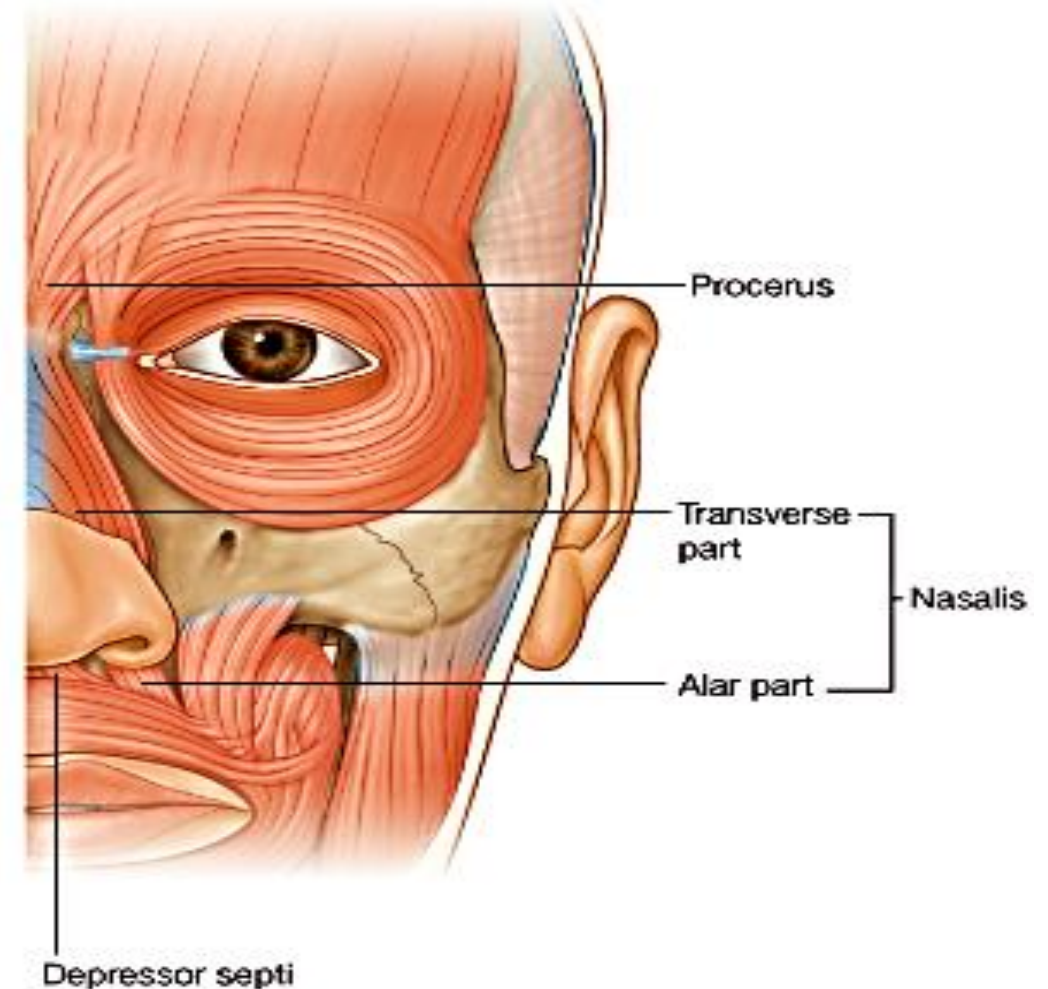


FACIAL MUSCLES

- **Nasal group:**

Three muscles are associated with the nasal group:

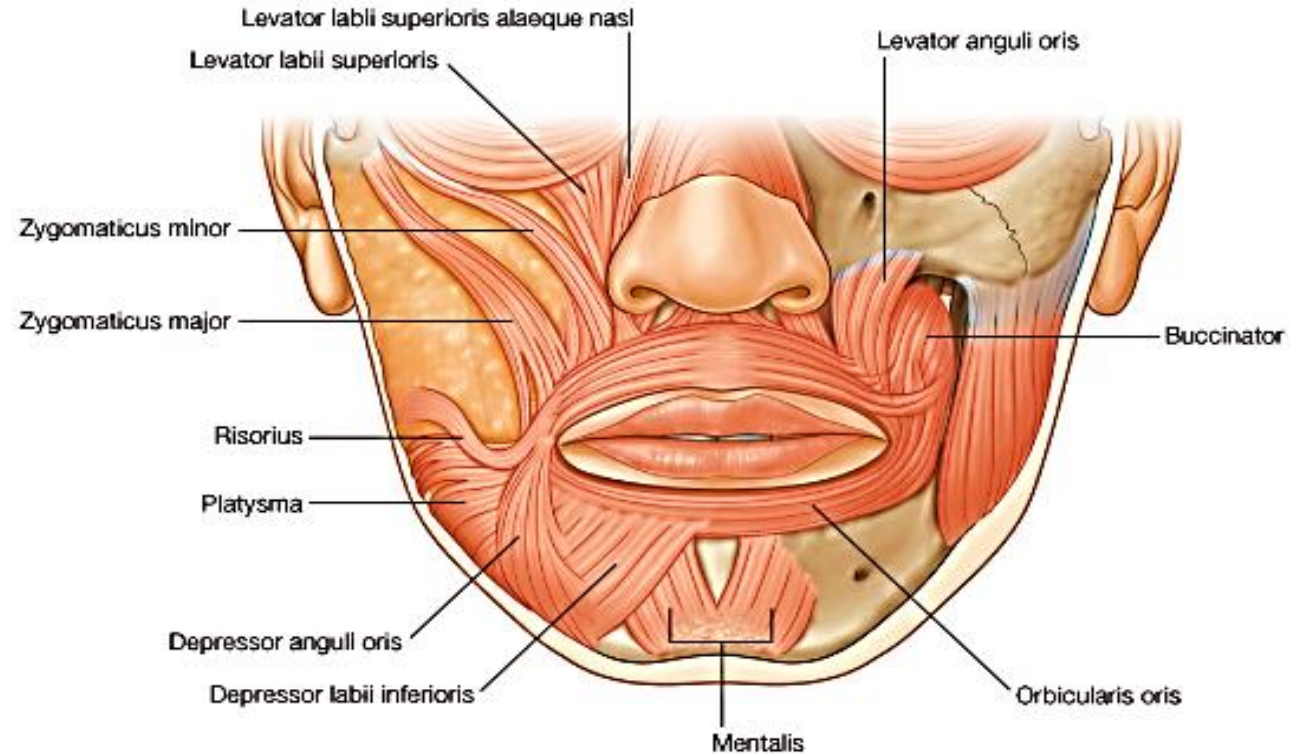
1. **Nasalis**
2. **Procerus**
3. **Depressor Septi Nasi**



FACIAL MUSCLES

- **Oral group:**

The muscles in the oral group move the lips and cheek. They include the **Orbicularis Oris** and **Buccinator** muscles, and a lower and upper group of muscles

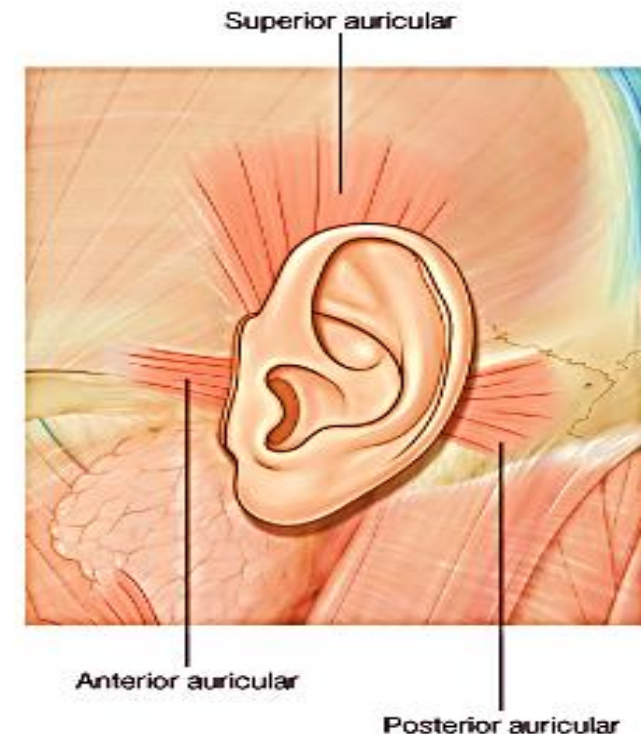


FACIAL MUSCLES

- **Auricular muscles:**

Three of these muscles, 'other muscles of facial expression', are associated with the ear-the anterior, superior, and posterior **auricular muscles**

1. Anterior muscle is anterolateral and pulls the ear upward and forward.
2. Superior muscle is superior and elevates the ear.
3. Posterior muscle is posterior and retracts and elevates the ear.



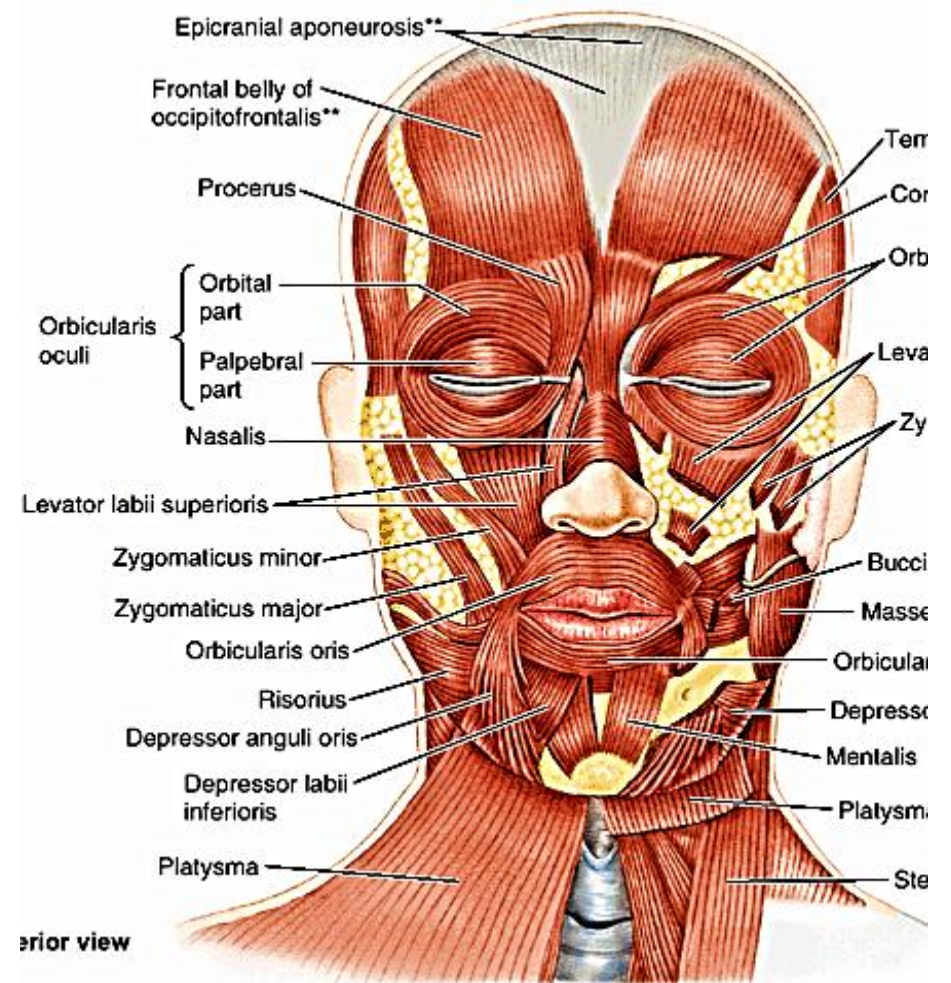
FACIAL MUSCLES

- **Platysma** :

is a large, thin sheet of muscle in the superficial fascia of the neck.

- **Occipitofrontalis** :

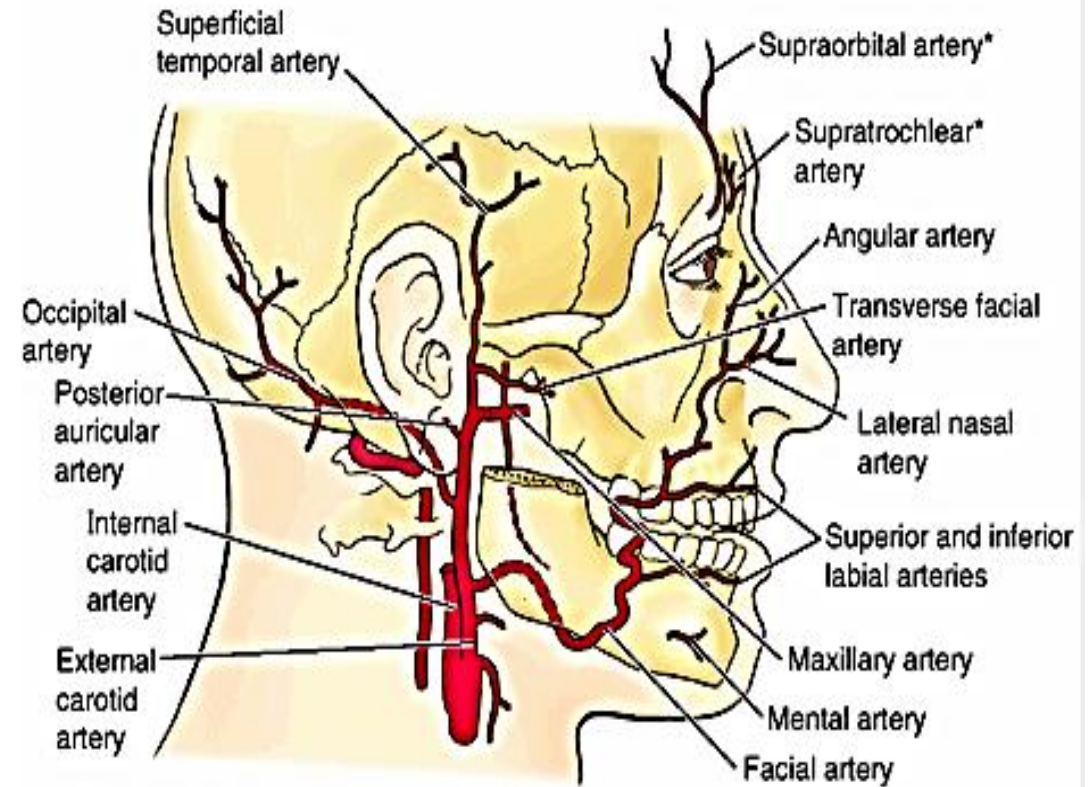
It consists of a frontal belly anteriorly and an occipital belly posteriorly.



Blood Supply of Face

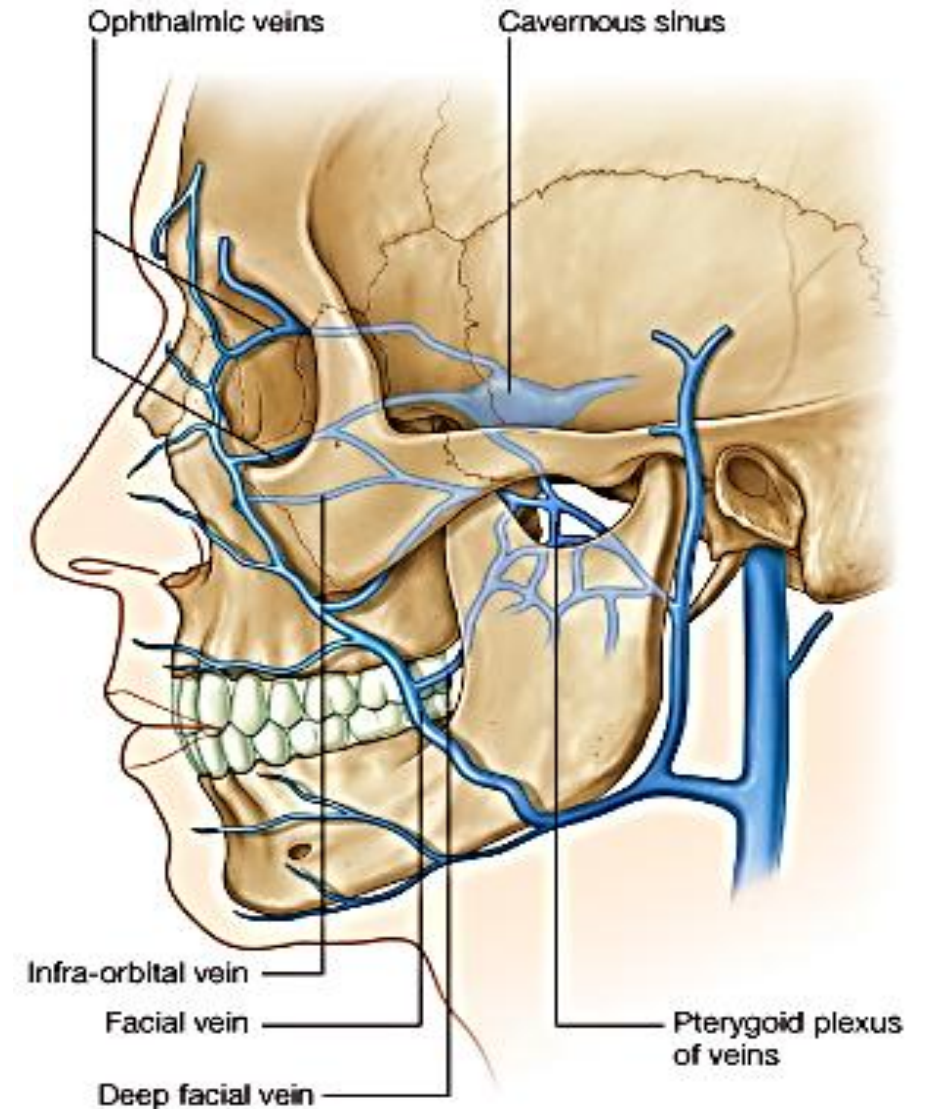
- Primarily from branches of the external carotid artery, though there is some limited supply from a branch of the internal carotid artery.

1. Facial A.
2. Transverse facial A.
3. Maxillary A.
4. Dorsal nasal A.



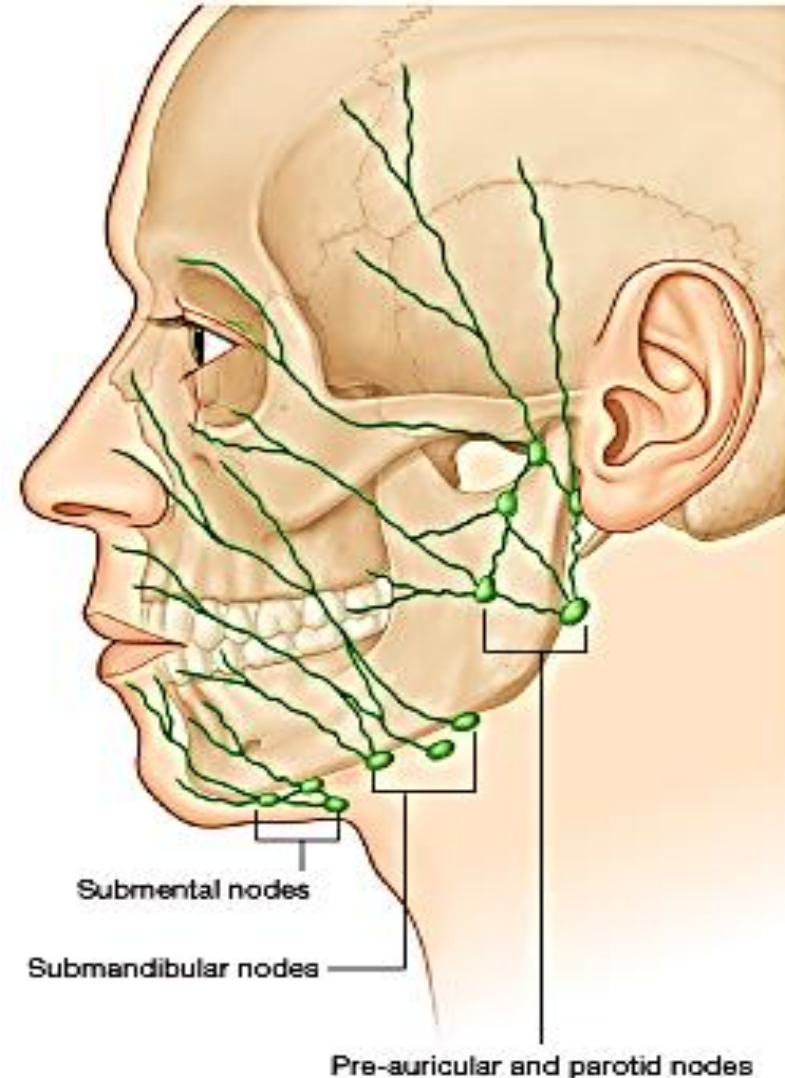
Venous drainage of Face

- Most of the venous return is back to the internal jugular vein, though some important connections from the face result in venous return through a clinically relevant intracranial pathway involving the cavernous sinus.



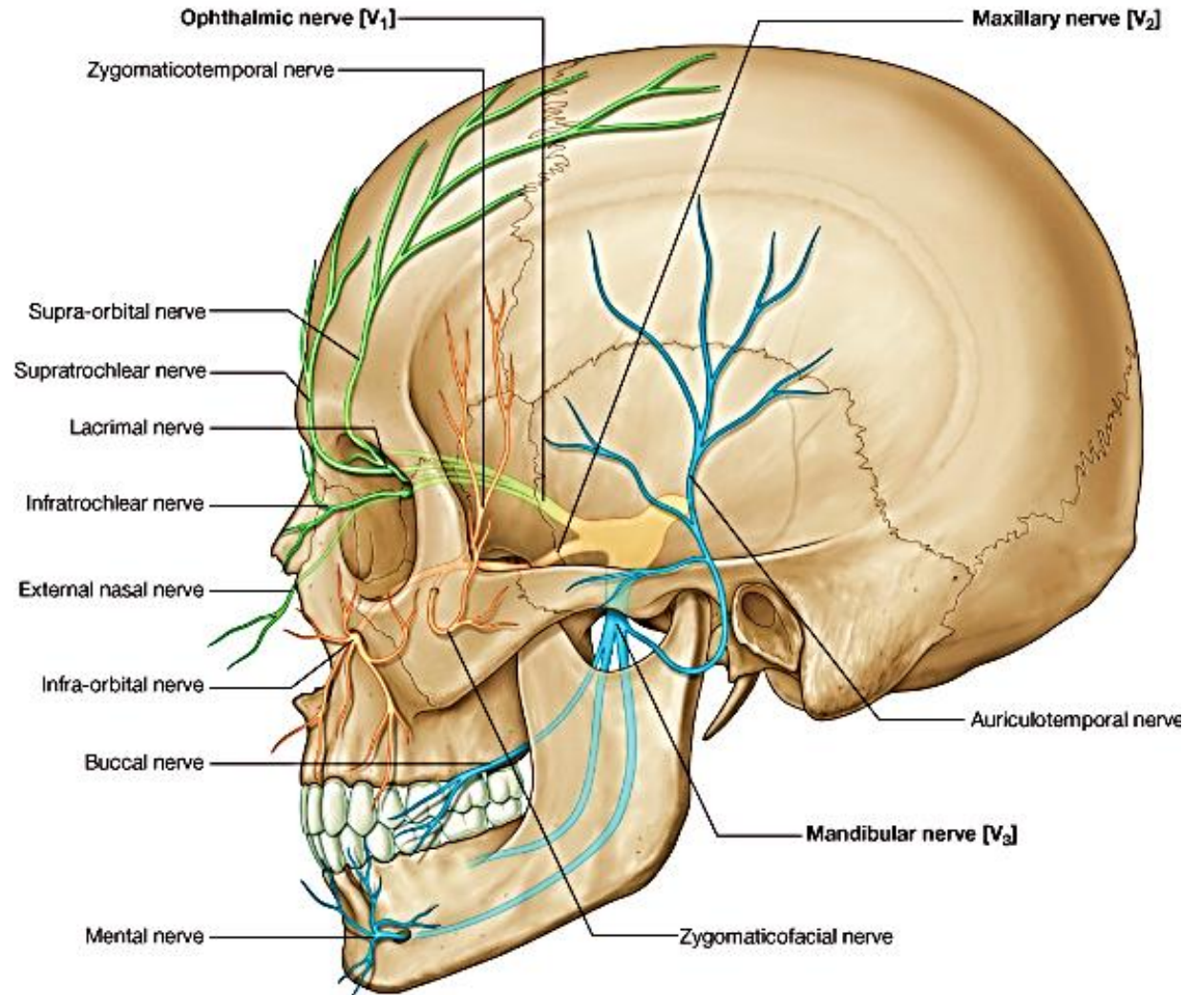
Lymphatics of Face

- Lymphatic drainage from the face primarily moves towards three groups of lymph nodes:
 1. Submental nodes
 2. Submandibular nodes
 3. Pre-auricular and parotid



Innervation of Face

- The cutaneous innervation of the face is by branches of the trigeminal nerve (V).
- a) Ophthalmic (V₁)
- b) Maxillary (V₂)
- c) Mandibular (V₃)



Paralysis of Facial Muscles

- Injury to the facial nerve or its branches produces paralysis of some or all facial muscles on the affected side (Bell's palsy).
 - i. Sad Face
 - ii. Smooth forehead
 - iii. Everted inferior eye lid
 - iv. Corneal dehydration and scarring
 - v. Drooping of mouth
 - vi. Impaired speech

